Document Control Report

Management of Patients taking Oral Anticoagulants in Community Hospitals Standard Operating Procedure

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1.0	January 2013	Final	Ratified Drugs and Therapeutics Committee on 17 January 2013
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2.0	January 2013	Final	Final version with minor amendments. Original held with the Medicines Management Team – Eastern.
2.1	October 2014	Draft	Updated after discussion with Karena Mulcock following addition of further Novel Oral Anti- coagulants and changes in product licenses
2.2	January 2014	Draft	Final draft incorporating further comments from Karena Mulcock and Jason Pruchniewicz and Drugs and Therapeutic Committee
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Contents

Document Control Report 1
1 Introduction and version history 4
2 Purpose 5
3 Scope 5
4 Location 5
5 Equipment 5
6 Procedure 6
7 References 14
8 Associated Documentation 15
9 Glossary of Terms 16

1 Introduction and version history

1.1 Introduction

There are a number of oral anticoagulants currently available on the UK market. These include the Vitamin K antagonists (VKA's): warfarin, phenindione, acenocoumarol, and the Novel oral anti-coagulants (NOAC's): dabigatran, rivaroxaban, and apixaban. Anticoagulants are considered as 'high risk' medication for monitoring and reporting of incidents, omitted doses and transfer of information between care providers.

Warfarin is currently the most commonly used oral anticoagulant in the UK. It is licensed for prophylaxis of embolisation in rheumatic heart disease and atrial fibrillation (AF); prophylaxis after insertion of prosthetic heart valve; prophylaxis and treatment of venous thrombosis and pulmonary embolism (VTE); transient ischaemic attacks (TIA).

At the time of writing the NOAC's:Dabigatran, Rivaroxaban and Apixaban are all licensed for the following indications: Prevention of stroke and systemic embolism in adult patients with non-valvular AF in patients with one or more specified risk factors, prophylaxis of venous thromboembolism in adults after total hip or total knee replacement surgery, treatment of deep vein thrombosis (DVT) and pulmonary embolism (PE), and prevention of recurrent DVT and PE in adults.

Acenocoumarol and Phenindione may be used in patients where warfarin or the NOAC's are contraindicated. For advice on the prescribing of Acenocoumarol and Phenindione please contact the haematology department at the acute trust.

Oral anticoagulation is monitored using the International Normalised Ratio (INR). This can be measured using a Point of care finger prick test (PoCT) or using a venous blood sample. INR must be monitored regularly in patients prescribed warfarin. INR monitoring is not required and is of no benefit in patients prescribed NOAC's.

The main complication of oral anticoagulant therapy is bleeding. Clinicians and patients need to be aware of this risk and how to assess and manage bleeding if it occurs.

It is important that clinicians and patients are aware of their responsibilities in the use and monitoring of oral anticoagulants, plus the appropriate methods of managing over anticoagulation.

All staff caring for patients on oral anticoagulation therapy must have the necessary work competencies .Any gaps in competence must be addressed through training to ensure that all staff may undertake their duties safely.

1.2 Version History – NHS Devon Provider Services

Version	Date Issued	Brief Summary of Change	Owner's Name
0.1	23/05/08	Initial draft document	Beth Atkins
0.2	19/06/08	Amendments after comment from Modern Matrons 1.2 and 3.10	Beth Atkins
0.3 final	24/06/08	Verified by John Coop	
v 1.1	08/05/09	Reviewed and amended 1.2	Karena Mulcock / Jane Watson
V2.0	20/05/09	Ratified at Professional Council	Karena Mulcock / Jane Watson

2 Purpose

The Standard Operating Procedure (SOP) has been written to inform staff about the appropriate management of patients taking oral anticoagulants in Community Hospitals.

3 Scope

This Standard Operating Procedure (SOP) relates to the following staff groups who may be involved in the management of patients taking oral anticoagulants in Community Hospitals:

- Registered nurses
- Medical staff
- Pharmacy staff

4 Location

This Standard Operating Procedure 'Management of Patients taking oral anticoagulants in Community Hospitals' can be implemented in all clinical areas where competent staff are available to undertake this role.

Staff undertaking this procedure must be able to demonstrate continued competence as per the organisations policy on assessing and maintaining competence.

5 Equipment

Point of Care Testing (PoCT) equipment if available.

6 Procedure

6.1 Admitting Patients taking Oral anti- coagulants

6.1.1 Patients admitted to community hospitals on oral anticoagulants should have a thorough drug history taken to include any over-the-counter herbal or alternative remedies taken. Any allergies should be clearly documented. This is part of the medicines reconciliation process.

6.1.2 An accurate history of anticoagulant dosing prior to admission should be obtained to inform subsequent dosing during admission. This may be obtained from:

- Patient and/or carer/relative interview
- NPSA 'Oral Anticoagulant Therapy' Information booklet for patients ('Yellow Book')
- Patients anticoagulant dosing service or GP (as applicable)
- Patient's transfer documentation
- Patient's prescription chart from a referring hospital

6.1.3 Patients taking warfarin should have a blood sample taken for an INR test within 24 hours of admission.

6.1.4 The patient's usual anticoagulant dosing service or GP should be informed of the patient's admission to hospital at the earliest opportunity.

6.2 Initiating Oral Anticoagulants In Community Hospitals

6.2.1 The doctor with the duty of care for the patient during admission is responsible for initiating any oral anticoagulation in the inpatient setting.

6.2.2 The risks and benefit of oral anticoagulant treatment should be considered and discussed with the patient/carers prior to commencement.

6.2.3 Prior to initiating treatment, the appropriateness of oral anticoagulation should be considered in light of the patient's current health, social status, co- morbidity, concomitant treatment and preferences. Consideration should be given to the patient's capacity to concord with treatment on discharge.

6.2.4 On initiating oral anticoagulant treatment, anticoagulant treatment booklets should be issued to all patients and completed.; these booklets include advice for patients on anticoagulant treatment, an alert card to be carried by the patient at all times, and a section for recording of INR results (applicable to warfarin) and dosage information. See section 6.3 for full details.

6.2.5 For information on switching between oral anti-coagulants, consult product literature.

Initiating warfarin

6.2.6 Warfarin doses should be prescribed on the oral anticoagulant record section of the prescription chart (PMAR). It should also be written inside the main chart under regular prescriptions, with the dose "as per chart".

6.2.7 The indication and target INR should be completed on the Oral Anticoagulant record in the PMAR.

6.2.8 An INR should be taken prior to initiating warfarin therapy to provide a baseline and inform the loading schedule. Liver function tests and urea and electrolytes are also recommended.

6.2.9 The loading schedule for oral anticoagulation with warfarin will be dependent on the indication for treatment and the patients' likely sensitivity to warfarin but should be in accordance with the British Committee for Standards in Haematology (BCSH) and local formulary guidance.

6.2.10 A slow induction schedule should be considered for patients with atrial fibrillation. Please refer to the local formulary for details:

http://northeast.devonformularyguidance.nhs.uk/formulary/chapters/2.-cardiovascular/anticoagulation/warfarin-initiation-and-inr-targets

6.2.11 Where there is an immediate risk of thromboembolus concurrent low molecular weight heparin (LMWH) should be prescribed to provide anticoagulant cover for at least 5 days and until a therapeutic INR has been achieved for two consecutive days. For further information, refer to the Clinical Guideline for the use and monitoring of low molecular weight heparins (LMWHs) in community hospital and community settings.

6.2.12 Target INR value and duration of treatment will be dependent on the indication for treatment but should be in accordance with the British Committee for Standards in Haematology and local formulary guidance. http://northeast.devonformularyguidance.nhs.uk/formulary/chapters/2.-cardiovascular/anticoagulation/warfarin-initiation-and-inr-targets

6.2.13 Where applicable the patient should be referred to the Anticoagulant dosing service that will be responsible for the on-going management of the patient's anticoagulant therapy during admission and/or on discharge.

Initiating NOAC's

6.2.14 NOAC's should be prescribed inside the main patient drug chart under regular prescriptions and annotated with the words "Oral Anticoagulant"

6.2.15 INR monitoring is not required for patients on NOAC's but renal function must be checked prior to initiation.

6.2.16 The dose of NOAC's varies depending on the indication for treatment, patient's age and renal function. Refer to the current BNF and local formulary guidance for prescribing and dosing information for individual drugs.

6.2.17 For information on switching between warfarin and NOAC's consult product literature.

6.2.18 When switching from LMWH, NOAC's should be started 0-2 hours prior to the time that the next dose of LMWH is due.

6.2.19 Duration of treatment will be dependent on the indication for treatment but should be in accordance with the current BNF and local formulary guidance.

6.3 Patient Information- verbal and written communication

6.3.1 Prior to the administration of anticoagulation, patients should receive adequate verbal and written information regarding their anticoagulant therapy. The verbal information should include as a minimum:

- A definition of anticoagulation.
- The indication for and the anticipated duration of treatment.
- The need for regular INR blood test monitoring for warfarin
- That warfarin doses are made up of different strength (and coloured) tablets.
- The risk of haemorrhage especially if poorly monitored.
- The need to report excessive bleeding or bruising.
- The need to keep the anticoagulation record safe and up to date with dosage and INR results (if applicable)
- The need for caution when stopping or starting new drugs or OTC medications.
- The potential interactions with drugs, OTC preparations, alcohol and foodstuffs.

This information should be repeated during the hospital stay, and at the time of discharge, and the patient given an opportunity to ask any questions regarding their anticoagulant treatment.

Extra care should be taken with patients with cognitive dysfunction, or those who have reduced hearing or sight, as they may receive substandard counselling. This may involve more prolonged conversations, as well as early communication with the patient's carers, family, and the GP.

6.3.2 Patients should be taken through the anticoagulant information booklet by a member of medical, nursing or pharmaceutical staff with the opportunity to ask any questions regarding the treatment.

The following sections in the NPSA yellow booklet should be fully completed by staff in liaison with the patient and/or carer:

- Patient's details
- Contact Numbers
- Condition requiring oral anticoagulant treatment
- Oral anticoagulant prescribed
- Target therapeutic INR range for Warfarin
- Intended duration of treatment

6.3.3 Although the NPSA Yellow booklet focuses on treatment with warfarin, it can still be used to record details of the patient's treatment with a NOAC.

6.3.4 Community hospitals can order the NPSA booklet from WEB basket. (OAT pack Cat UOI: pack of 50 sets.

Electronic copies of the booklets in a selection of languages, and further advice are available at: www.npsa.nhs.uk/nrls/alerts-and-directives/alerts/anticoagulant.

Patient information leaflets and alert cards for the individual NOAC's are also available from:

Dabigatran: http://www.pradaxa.co.uk/downloads/patient-alert-card.pdf

Rivaroxaban:http://www.xareltoinfo.co.uk/static/documents/pdf/Patient_Alert_Card_15and20mg_nov2012.pdf

Apixaban:https://www.medicines.org.uk/media/3705/19546340_eliquis_patient_alert_card_web_uk_final-1-.pdf

6.4 Managing Oral Anticoagulant Therapy in Community Hospitals

Warfarin

6.4.1 Warfarin is available in four tablet strengths so as to avoid the need to halve tablets. Warfarin is manufactured in 5mg (pink), 3mg (blue), 1mg (brown) and 500microgram (white) tablets.

6.4.2 Warfarin should be taken at the same time each day. IIt is taken at 6pm unless otherwise specified to allow for appropriate morning INR monitoring and subsequent dose adjustment. If the dose is prescribed and administered at a different time, the dose must be administered as prescribed.

6.4.3 If a dose of warfarin is missed refer to the Omitted and Delayed medicines SOP. No double doses should be taken.

6.4.4 If warfarin therapy is to be stopped then it can be stopped suddenly without any complications. Inpatients should be assessed for alternative thromboprophylaxis once their INR is no longer in therapeutic range.

6.4.5 Inpatients should have their anticoagulation status, INR, monitored within 24 hours of admission and subsequently as directed by the dosing clinician.

6.4.6 Hospital patients will have a greater need for closer monitoring due to the effect that illness, drugs and changes in diet will have on the INR. A suggested monitoring schedule is:

- Daily INRs for very ill patients including those started on antibiotics, steroids and other medications known to interact with warfarin.
- Close monitoring of patient with interactions disease states such as those with liver or renal disease.
- For stable patients having rehabilitation INR monitoring can be done no less frequently than every seven days.

6.4.7 When INR monitoring is clinically not indicated such as patients requiring an end of life (EoL) care pathway or where the clinical condition of the patient would mean that interpretation of the INR results would be compromised then this should be documented and a decision made as to the value of continuing both anticoagulant treatment and monitoring.

6.4.8 It is the responsibility of the administering nurse to ensure that INR tests have been taken in accordance with the prescriber's instructions and that where an INR test has been taken that day the prescriber has reviewed the result, and prescribed in accordance with the result, before any subsequent administrations.

6.4.9 If Point of care testing (PoCT) is used and an INR result above 4.0 obtained, a subsequent venous sample must be taken. PoCT values above 4.5 show increasingly poor correlation with laboratory and reference methods and should not be used as a basis for treatment dosing.

6.4.10 All INR results should be recorded in the anticoagulation section of the prescription chart and appropriately annotated if a venous sample has been used.

6.4.11 When paper copies of patient's venous INR results are received these should be signed by the patient's doctor and then filed in the patient's medical notes.

6.4.12 Warfarin doses should normally only be prescribed up until the date of the next INR test. The prescriber is then required to prescribe subsequent doses in response to the INR result. For patients stable on their warfarin dose, i.e. where tests are not indicated more frequently than once a week, it is reasonable to prescribe the dose up until the next working day post the INR test to allow the prescriber to come in and prescribe in person.

6.4.13 For remote prescribing of oral anticoagulants refer to Section 5.1.17 of the Medicines Policy.

6.4.14 Where a patient is managed in the community through a Point of care testing service this may continue under the responsibility and supervision of their GP during a community hospital inpatient stay. INR results should still be recorded and warfarin doses prescribed on the Oral Anticoagulant section of the prescription chart.

NOACs

6.4.15 NOACS should be taken at the same time each day. For information on forgotten or delayed doses please refer to the individual product literature. No double doses should be taken.

6.4.16 Renal function should be monitored annually and when necessary, for example, when other drugs with renal effects are started or altered, or with dehydration/vomiting/diarrhoea.

6.5 Perioperative Anticoagulant Management

6.5.1 Oral anticoagulant therapy may need to be interrupted peri-operatively to reduce the risk of intra-operative and post-operative bleeding.

6.5.2 The clinician responsible for dosing the patients' anticoagulant therapy and the surgical team should liaise to ensure action is taken to achieve appropriate peri-operative coagulation status.

6.6 Management of over anticoagulation and/or bleeding

6.6.1 The main adverse effect of all oral anticoagulants is haemorrhage. This is associated with a significant morbidity and mortality. The risk of haemorrhage is increased in the elderly, in those on interacting medication and in circumstances where the prescribing and monitoring of anticoagulation is inadequate.

Warfarin

6.6.2 All hospitals must stock phytomenadione (Vitamin K1). Refer to BNF section 5.2 for most up to date advice on managing over anticoagulation.

6.6.3 An INR of greater than 6 is considered to be over anticoagulation and action should be taken to address this. The objective being to return the INR to therapeutic levels in a timely manner but without exposing the patient to unnecessary treatment or rendering their anti-coagulation sub-therapeutic.

6.6.4 INR 5.0–8.0, no bleeding—withhold 1 or 2 doses of warfarin and reduce subsequent maintenance dose.

6.6.5 INR 5.0–8.0, minor bleeding—stop warfarin; give phytomenadione (vitamin K1) 1–3 mg by slow intravenous injection; restart warfarin when INR <5.0 Reduce subsequent maintenance dose.

6.6.6 INR greater than 8.0, no bleeding—stop warfarin; give phytomenadione (vitamin K_1) 1–5 mg by mouth using the intravenous preparation orally [unlicensed use]; Recheck INR the next day, repeat dose of phytomenadione if INR still too high after 24 hours. Restart warfarin when INR <5.0 (BNF).

6.6.7 INR greater than 8.0, minor bleeding—stop warfarin; give phytomenadione (vitamin K_1) 1–3 mg by slow intravenous injection; repeat dose of phytomenadione if INR still too high after 24 hours; restart warfarin when INR drops to below 5.0.

6.6.8 If there is significant but not life threatening bleeding, whatever the INR, stop the warfarin. Clinical judgement should be used to decide the best course of action. Vitamin K_1 (phytomenadione) 5mg by slow intravenous injection is generally adequate or transfer patient to nearest acute hospital.

6.6.9 If there is life threatening haemorrhage, dial 999 and have the patient transferred to the nearest acute hospital.

6.6.10 Intravenous vitamin K works in 4-8 hours while oral vitamin K 6-10 hours. IV vitamin K (Konakion MM) is mixed with 10ml 5% glucose and given over 3-5 minutes. Oral vitamin K is administered by drawing up the required volume of IV Konakion MM preparation in a syringe and diluting it with 5mls water, removing the needle and expelling the contents into the patient's mouth.

6.6.11 In all cases, review of the patient's history and medication should be undertaken to ascertain the reason for the raised INR. A DATIX form must be completed for any patient with an INR over 6.0.

NOACs

6.6.12 Currently there is no licensed antidote or guaranteed means of reversal for patients taking a NOAC who present with bleeding. Expert advice from Haematology should be sought.

6.6.13 Overdose can be treated with oral activated charcoal if within 2 hours of ingestion. Dabigatran can be dialysed.

6.6.14 Interruption of treatment may be sufficient to treat bleeding. The estimated time for restoration of haemostasis after stopping treatment in patients with adequate renal function is usually within 12 hours for dabigatran and apixaban and 24 hours for rivaroxaban. These times will be extended in patients with renal impairment.

6.6.15 Check FBC, U&Es and a coagulation screen (PT, Thrombin Time and APTT). If these tests are within the normal reference range it is likely that only a low level of the anticoagulant is present. Standard clotting tests cannot be used to estimate the extent of the anticoagulant effect. The anticoagulant effect of NOACs will be prolonged in renal impairment.

6.6.16 If there is life threatening haemorrhage, dial 999 and have the patient transferred to the nearest acute hospital.

6.6.17 In all cases, review of the patient's history and medication should be undertaken to ascertain the reason for the bleed. A DATIX form must be completed for any patient with a bleed whilst taking a NOAC.

6.7 Plans for Transfer or Discharge

6.7.1 Prior to discharge, a competent member of staff should re-iterate to the patient verbal information about their oral anticoagulant treatment. (see Section 6.3) The patient's anti- coagulation booklet should be completed and given to the patient or carer.

6.7.2 The patient must be advised of the daily anticoagulant dose they are to take.

6.73 Staff should ensure for patients taking warfarin, they have a clear understanding of the date their next INR test is due, how to obtain the results, and any subsequent instructions to amend their warfarin dose.

6.7.4 Patients should be advised that in the event of haemorrhage or significant acute illness to omit their anticoagulant medication and seek urgent medical advice.

6.7.5 The patient should be provided with an appropriate quantity of anticoagulant medication on discharge and advised how to obtain further supplies of medication.

6.7.6 The NPSA safety alert No 18 'Actions that can make anticoagulant therapy safer' advises that a risk assessment should be carried on the use of monitored dosage systems for warfarin in individual patients. The general use of monitored dosage systems for warfarin should be minimised as dosage changes using these systems are more difficult.

6.7.7 Dabigatran is not suitable for inclusion in a compliance aid or monitored dosage system (e.g. blister pack) as the capsules are moisture sensitive and should not be stored outside their packaging. Rivaroxaban and Apixaban can be included in a compliance aid if necessary.

6.7.8 The community hospital should ensure that the prescriber or service responsible for continuing the patient's anticoagulant management is provided with the necessary information and is informed of the time of the patient's discharge/transfer.

6.7.9 The discharge summary should be completed detailing the indication and duration of oral anticoagulation therapy, the target INR and recommended monitoring. A copy of the anticoagulation section of the prescription chart should also be attached if necessary.

6.8 Training and Education of staff initiating anticoagulation

It is essential that all staff caring for patients on oral anticoagulant therapy are skilled in the necessary competencies. All staff involved in anticoagulation prescribing, administration and monitoring should complete appropriate training. This may include:

E-learning modules available on the NPSA, MHRA, BMJ and local formulary websites:

- http://www.npsa.nhs.uk/health/alerts
- http://new-learning.bmjknowledge.org/learning/user/login.html
- http://www.mhra.gov.uk/ConferencesLearningCentre/LearningCentre/Medicineslearningmodules/Oralanticoagulants/index.htm
- http://northeast.devonformularyguidance.nhs.uk/Learning/Quizzes/anticoagulation_quiz.html

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Staff unsure of the appropriate action when prescribing or administering anticoagulant therapy should seek senior advice and consider their training requirements immediately.

7 References

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11. Peninsula Heart and Stroke Network, 2013.*Novel oral anticoagulants for the prevention of stroke and systemic embolism in atrial fibrillation* [Online] Available at: http://www.northeast.devonformularyguidance.nhs.uk/Chapters/02-Cardiovascular/Pen-NOAC-2013.pdf

12. Royal Devon & Exeter NHS Foundation Trust, 2011. *Reversal of Anticoagulation Therapy*

13. Royal Devon & Exeter NHS Foundation Trust, 2010. *Anticoagulation Policy.*

8 Associated Documentation

- Clinical Guideline for the use and monitoring of low molecular weight heparins (LMWHs) in community hospital and community settings
- Medicines Policy 2012
- SOP CH007 Medicines Reconciliation
- Point of care Testing for INR using Coagucheck SOP November 2011

9 Glossary of Terms

AF	Atrial Fibrillation
APTT	Activated partial thromboplastin time
BCSH	British Committee for Standards in Haematology
e-GFR	Electronic Glomerular Filtration Rate – a calculated estimation of renal function
INR	International Normalised Ratio – the test used to assess degree of anticoagulation with warfarin
LMWH	Low-molecular weight heparin – e.g. dalteparin
NPSA	National Patient Safety Agency
NOAC	Novel oral anticogaulant
PoCT	Point of Care Testing
PMAR	Prescription Medication and Administration Record (drug chart)
PT	Prothrombin time
TIA	Transient Ischaemic Attack
VKA	Vitamin K antagonist eg warfarin
VTE	Venous thrombo-embolism